Practice Problems
Calculus-Based Physics II

Chapter 38 Electric Potential

1) The units of Electric Potential energy are

A) J/Kg
B) J/C
C) J
D) J/N
E) N
F) N/C

ΔPE_E → Joules

2) The units of Electric Potential are

A) J/Kg
B) J/C
C) J
D) J/N
E) N
F) N/C

$1V = 1\ J/C$

3) Electric potential, measured in volts, is the ratio of electric potential energy to amount of electric

A) current.
B) charge.
C) resistance.
D) voltage.
E) None of the above choices are correct.

$\Delta V = \frac{\Delta PE}{q}$

4) If it requires 10 J of work to push an object with 2 Coulombs of net charge in a uniform electric field, what is the change in the potential of the object?

A) 5 V
B) 2 V
C) 10 V
D) 20 V
E) 0 V

$W = q\Delta V$

$\Delta V = \frac{W}{q} = 5\ V$

5) If it requires 10 J of work to push an object with 2 Coulombs of net charge in a uniform electric field, what is the change in the potential energy of the object?

A) - 10 J
B) - 20 J
C) - 5 J
D) - 2 J
E) 0 J

$\Delta PE = -W$

6) How much work is needed to carry an electron from the positive terminal to the negative terminal of a 9.0-V battery. ($e = 1.60 \times 10^{-19}$ C , $m_{electron} = 9.11 \times 10^{-31}$ kg)

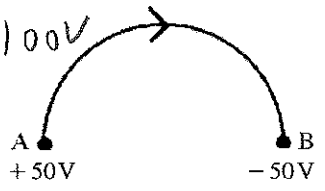
$W = q\Delta V = 1.44 \times 10^{-18}$ J

7) How much work must we do on an electron to move it from point A, which is at a potential of +50V, to point B, which is at a potential of -50 V, along the semicircular path (r = 10 cm) shown in the figure? Assume the system is isolated from outside forces. ($q = 1.60 \times 10^{-19}$ C)

$W = q\Delta V$

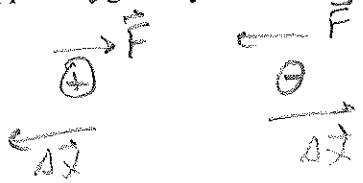
$W = -1.6 \times 10^{-17}$ J

$\Delta V = V_f - V_i = -100V$



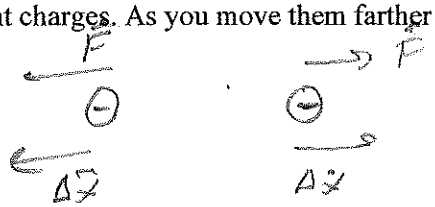
8) Suppose you have two point charges of opposite sign. As you move them farther and farther apart, the potential energy of this system relative to infinity

(A) increases.
B) decreases.
C) stays the same.



9) Suppose you have two negative point charges. As you move them farther and farther apart, the potential energy of this system relative to infinity

A) increases.
(B) decreases.
C) stays the same.



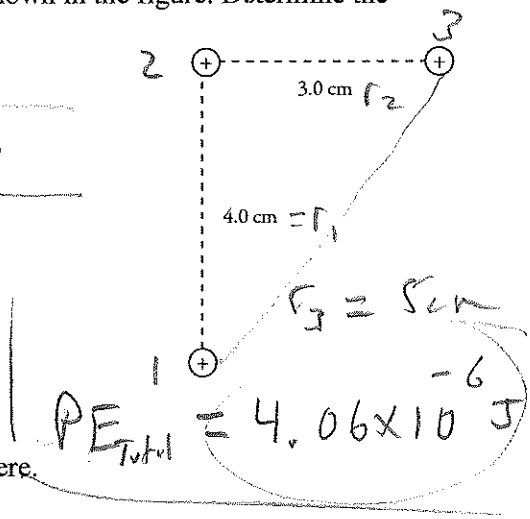
10) Consider the group of three positively charged point charges q = 2.4 nC shown in the figure. Determine the electric potential energy of this system of charges relative to infinity?

$PE_{Total} = PE_{1st} + PE_{2nd} + PE_{3rd}$

$PE_{1st} = \frac{K q_1 (0)}{r_2} = 0$ no other charges

$PE_{2nd} = \frac{K q_1 q_2}{r_1} = 1.3 \times 10^{-6}$ J

$PE_{3rd} = \frac{K q_1 q_3}{r_3} + \frac{K q_2 q_3}{r_2} = 2.76 \times 10^{-6}$ J



$PE_{Total} = 4.06 \times 10^{-6}$ J

11) When the electric field is zero at a point, the potential must also be zero there.

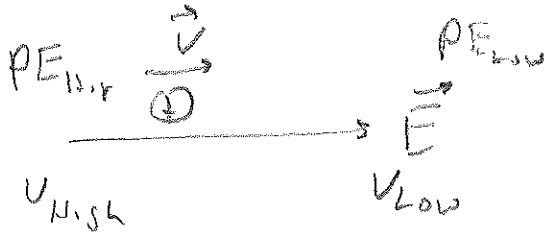
A) True
(B) False

12) If the electrical potential in a region is constant, the electric field must be zero everywhere in that region.

(A) True
B) False

$\Delta V = E \Delta x = 0$

13) For a proton moving in the direction of the electric field



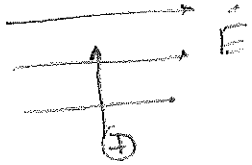
A) its potential energy increases and its electric potential decreases.
B) its potential energy decreases and its electric potential increases.
(C) its potential energy decreases and its electric potential decreases.
D) its potential energy increases and its electric potential increases.
E) both its potential energy and it electric potential remain constant.

14) A negative charge, if free to move in an electric field, will move



A) from high potential to low potential.
B) in the direction of the electric field.
C) away from infinity.
(D) from low potential to high potential.
E) toward infinity.

15) As a proton moves in a direction perpendicular to the electric field lines

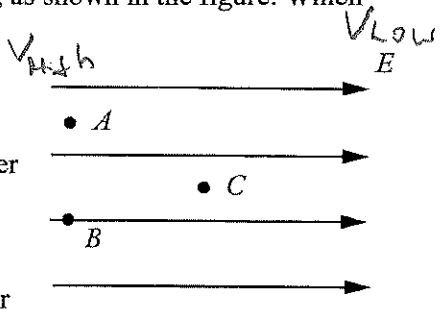


A) it is moving from low potential to high potential and gaining electric potential energy.
B) it is moving from low potential to high potential and losing electric potential energy.
C) it is moving from high potential to low potential and gaining electric potential energy.
D) it is moving from high potential to low potential and losing electric potential energy.
(E) both its electric potential and electric potential energy remain constant.

16) If the electric potential at a point in space is zero, then the electric field at that point must be

A) negative.
B) zero.
C) uniform.
D) positive.
(E) impossible to determine based on the information given.

17) Suppose a region of space has a uniform electric field, directed towards the right, as shown in the figure. Which statement about the electric potential is true?

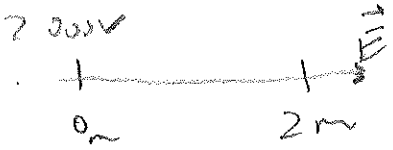


A) The potential at all three locations (A, B, C) is the same because the field is uniform.
B) The potential at points A and B are equal, and the potential at point C is higher than the potential at point A.
C) The potential at point A is the highest, the potential at point B is the second highest, and the potential at point C is the lowest.
(D) The potential at points A and B are equal, and the potential at point C is lower than the potential at point A.

18) At a certain point in space there is a potential of 400 V. What is the potential energy of a +2-μC charge at that point in space?

$\Delta PE = q\Delta V = 8 \times 10^{-4}$ J

19) An 800 V/m electric field is directed along the +x-axis. If the potential at $x = 0$ m is 2000 V, what is the potential at $x = 2$ m?



$\Delta V = -|\vec{E}|\Delta x$

$\Delta V = -1600V$

$\Delta V = V_f - V_i$

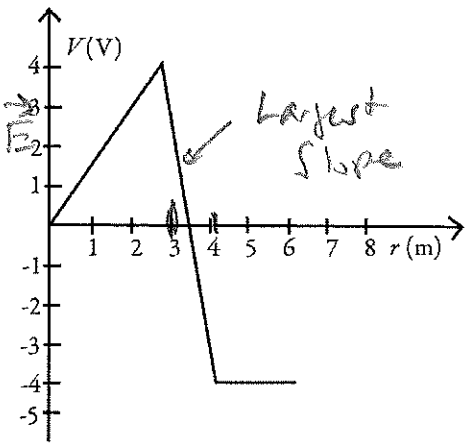
$V_f = V_i + \Delta V = 400V$

20) The graph in the figure shows the variation of the electric potential V (measured in volts) as a function of the radial direction r (measured in meters). For which range or value of r is the magnitude of the electric field the largest?

Slope $= \frac{\text{rise}}{\text{run}} = \frac{\Delta V}{\Delta x} = |E|$

A) from $r = 3$ m to $r = 4$ m
B) at $r = 3$ m
C) from $r = 4$ m to $r = 6$ m
D) from $r = 0$ m to $r = 3$ m
E) at $r = 4$ m

Largest Slope = Largest $|\vec{E}|$

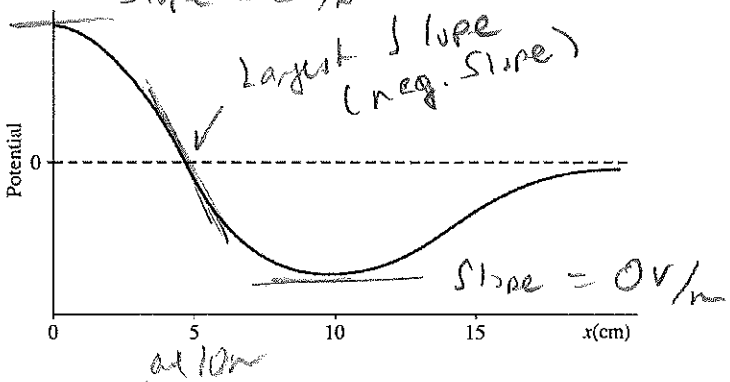


21) From the graph of V vs r above, what is the approximate electric field strength in the region of 0 to 3 meters?

$|\vec{E}| = \frac{\Delta V}{\Delta x} = \frac{4V - 0V}{3m - 0m} = 1.33\ V/m$

22) The potential as a function of position x is shown in the graph in the figure. Which statement about the electric field is true?

$\vec{E} = -\left(\frac{dV}{dx}\right)\hat{x}$



A) The electric field is zero at $x = 0$, its magnitude is at a maximum at $x = 5$ cm, and the field is directed to the right there.
B) The electric field is zero at $x = 5$ cm, its magnitude is at a maximum at $x = 0$, and the field is directed to the right there.
C) The electric field is zero at $x = 0$, its magnitude is at a maximum at $x = 15$ cm, and the field is directed to the left there.
D) The electric field is zero at $x = 10$ cm, its magnitude is at a maximum at $x = 5$ cm, and the field is directed to the left there.

0m +

23) In a certain region, the electric potential due to a charge distribution is given along the x-axis as

$$V(x) = ((2\ V/m)x - ((1\ V/m^2)\,x^2)$$

A) Determine the electric field at x = 0 m along the x-axis.

$\vec{E} = -\left(\frac{dV}{dx}\right)\hat{x} = -\left(2\ V/m - (2\ V/m^2)\,x\right)\hat{x}$

$\vec{E}(0m) = -(2\ V/m)\,\hat{x}$

B) Determine the electric field at x = 1 m along the x-axis.

$\vec{E}(1m) = -\left(2\ V/m - (2\ V/m^2)\,x\right)\hat{x}$

$\vec{E}(1m) = (0\ V/m)\,\hat{x}$

Answers

1) C
2) B
3) B
4) A
5) A
6) 1.44×10^{-18} J
7) 1.6×10^{-17} J
8) A
9) B
10) 4.06×10^{-6} J
11) B
12) A
13) C
14) D
15) E
16) E
17) D
18) 8×10^{-4} J
19) 400 V
20) A
21) 1.33 V/m
22) A
23) A) $(2\ \text{V/m})\hat{x}$ B) $(0\ \text{V/m})\hat{x}$